

Numpy

array_3d[0:2, 1, 1:]

$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 3 & 4 & 5 \end{bmatrix}$

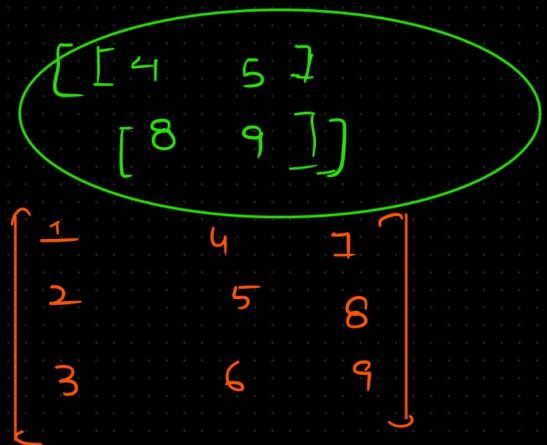
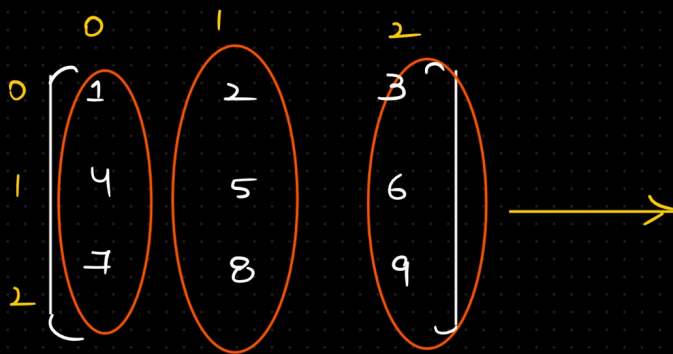
$\begin{bmatrix} 0 & 5 & 6 & 7 \\ 1 & 7 & 8 & 9 \end{bmatrix}$

array([[[1, 2, 3],
[3, 4, 5]],

[[5, 6, 7],
[7, 8, 9]],

[[9, 10, 11],
[11, 12, 13]]])

$\begin{matrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 7 & 8 & 9 \end{matrix}$



$\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \text{array}([2, 1, 3, 5, 4, 10, 7, 6]) \end{matrix}$

index $\rightarrow$ $\begin{matrix} 1 & 0 & 2 & 4 & 3 & 7 & 6 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 10 \end{matrix}$ 7 = 8

number

arr 1 arr 2

$\begin{matrix} 0 \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix} \\ 1 \rightarrow \begin{bmatrix} 4 & 5 & 6 \\ 5 & 6 & 7 \end{bmatrix} \end{matrix}$

axis = 0

$\left\{ \begin{matrix} [1 & 2 & 3] \\ [3 & 4 & 5] \\ [4 & 5 & 6] \\ [5 & 6 & 7] \end{matrix} \right\}$

$\left\{ \begin{array}{l} [1\ 2\ 3\ 4\ 5\ 6] \\ [3\ 4\ 5\ 5\ 6\ 7] \end{array} \right\}$

axis = 1

Identity Matrix $\rightarrow$ Diagonal

elements = 1

Internal code working

Transpose ✓

Sort $\rightarrow$

QuickSort, MergeSort,

HeapSort

Heap

Matrix Multiplication ✓

Divide & Conquer